

Pundarikaksh N. Tripathi

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Experience

FlyRank AI | Causal ML, X-Learner, Random Forest, scikit-learn, DuckDB, Hugging Face Datasets, Python

Remote

Machine Learning Engineering Intern

July 2026 – September 2026

- Built a refresh-opportunity ranker on 30,000 anonymized pages: dropped label-derived columns (`trend_pct`, `trend_direction`), trained a Random Forest on a client-grouped 70/30 split, raising Precision@50 from 0.56 (hand-written rule) to 0.70.
- Quantified leakage: random-split P@50 0.94 vs 0.70 under client-grouped holdout (0.24 optimism gap); reframed claims as observational.
- Capstone (mentor-accepted): X-Learner estimating the CATE of a content refresh (random-forest learners, decile Qini on a client-grouped holdout); ran DuckDB `hf://` Parquet queries over the full 79.8M-row warehouse.
- Converted scores into an action playbook (reason codes, 6,046 flagged pages, human-review no-go list); wrote a causal-ML landscape note (S/T/X-learners, DML, causal forests, Dragonnet, CEVAE, DeepIV, Qini) stating the unobserved-confounding limit and the need for an A/B test.

Projects

🔗 **Causal-DML: Counterfactual Churn Engine on KKBox Logs** | Python, DuckDB, EconML, DoWhy, FastAPI, MLflow, Double ML, Streamlit

- Aggregated the 30.5 GB KKBox `user_logs.csv` into per-user features with one in-process DuckDB query (`read_csv_auto`, GROUP BY), no cluster.
- Estimated heterogeneous effects with DoWhy and EconML LinearDML: random-forest nuisances, 3-fold cross-fitting, Neyman-orthogonal residual regression on 100k users, tracked in MLflow.
- Validated with a placebo design (synthetic confounded treatment on real churn labels, true effect zero): ATE -0.0017 , outcome R^2 0.006, propensity AUC 0.51.
- Served CATE via a FastAPI `/predict_cate` endpoint and a Streamlit/Plotly dashboard; API tests and CI (ruff, pytest, image builds).

🔗 [Source: github.com/PundarikakshNTripathi/Causal-DML](https://github.com/PundarikakshNTripathi/Causal-DML)

🔗 **HiveTorch: Federated Learning over gRPC and Kubernetes** | PyTorch, gRPC, protobuf, safetensors, Kubernetes, Prometheus, FedAvg

- Measured FedAvg on MNIST (20 clients, 3 seeds): 97.3% IID vs 96.4% at Dirichlet $\alpha=1$, 92.9% at $\alpha=0.1$ and 90.0% with 2-class shards.
- Implemented FedAvg in-process and as a gRPC federation (protobuf, safetensors weights, server-side validation of session, round and tensor shapes); the networked run reproduces the in-process accuracy trace to $1e-6$.
- Deployed as Docker Compose services and a Kubernetes Indexed Job (pod index picks the data partition) with Prometheus metrics and alerts; ran 1 server + 5 clients end to end on Minikube.
- Wrote the Dirichlet label-skew partitioner with tests (disjoint cover, determinism, entropy ordering); CI lints, type-checks and runs the container federation.

🔗 [Source: github.com/PundarikakshNTripathi/HiveTorch](https://github.com/PundarikakshNTripathi/HiveTorch)

🔗 **Aegis: Behavioral Governance for Autonomous Agents (OpenAI Buildweek)** | Go, C, eBPF LSM, seccomp, AWS Cedar, SQLite, LinUCB, RL

- Built stateful trajectory evals: an adaptive multi-step adversary against the real scorer, memory and adjudicator in simulation (mean 9.2 steps before denial, 10% over-refusal on legitimate workflows).
- Scoring costs $3.8 \mu\text{s/event}$; float32-BLOB embeddings cut episodic recall from 278 ms to 11 ms at 5,000 cases; 57 Go tests under the race detector; CI compiles the eBPF objects and gates the evals.
- Validated live on Linux and WSL2 in a recorded demo: blocked the post-exploitation payload of BunnyHijack (a disputed Bun package-manager PoC) at the syscall layer, plus other unauthorised commands.

🔗 [Source: github.com/Quiet-Intelligence/aegis](https://github.com/Quiet-Intelligence/aegis)

🔗 **nanoDist: Data Parallelism and ZeRO-2 from Scratch** | Python, NumPy, TCP sockets, ring all-reduce, ZeRO-2, FP16, Hydra, Optuna, W&B

- Built ZeRO stage 2 on a from-scratch ring all-reduce: gradient reduce-scatter, Adam on each rank's $1/N$ parameter shard with $1/N$ optimizer state, parameter all-gather; matches single-process Adam (2 and 4 processes).
- Added FP16 compute on FP32 master weights with loss scaling, gradient accumulation and activation checkpointing; analytic accounting for a $1024 \rightarrow 4096 \rightarrow 1024$ MLP: 105.97 $\rightarrow$ 33.84 MB per worker (-68%).
- 27 tests (Adam equivalence, finite-difference gradients, multi-process TCP); they exposed a crash in the checkpointing path, which I fixed; CI runs ruff, mypy and pytest.

🔗 [Source: github.com/PundarikakshNTripathi/nanoDist](https://github.com/PundarikakshNTripathi/nanoDist)

Achievements

- **3rd on the public and private leaderboards (89,000+ teams), Amazon ML Challenge 2026** at the last check; macro-F0.5 0.991762. Led planning, coordination and review of a 3-person team's entity-resolution pipeline (1.73M businesses vs ≈ 10 M noisy records): 13 lexical retrievers + fine-tuned e5 kNN (99.82% candidate recall), LightGBM $A \rightarrow B \rightarrow C$ cascade (87 features, cross-encoder blend, 5-seed bag), calibration and expected-F0.5 set selection.
 - Diagnosed a validation/leaderboard gap (0.9896 vs 0.981) as unseen-country shift (France appears only in test; global IDF gave $\approx 42\%$ of French tokens maximal weight); fixed with per-country IDF and pseudo-labels, adopting changes only if leave-one-country-out improved both ways (US $\rightarrow$ India 0.936 $\rightarrow$ 0.952).
- **2nd place, Masai $\times$ NoBroker HackArena 2025**: CivicAgent, an AI civic-complaint triage platform (Gemini Vision classification, APScheduler SLA tracking with email escalation, Supabase/PostGIS); proof of concept, no longer hosted.

Technical Skills

Languages: Python, C++20, CUDA C++, PTX (inline asm), Go, C, SQL, TypeScript, Bash

ML Frameworks: PyTorch, JAX, Transformers (Hugging Face), ONNX, Triton, scikit-learn, LightGBM, EconML, DoWhy

Data and Backend: DuckDB, PostgreSQL, SQLite, FastAPI, Supabase, MLflow, Prometheus

Education

Babu Banarasi Das University

Lucknow, UP

B.Tech, Computer Science and Engineering (Artificial Intelligence)

Expected August 2027

- **Coursework:** Probability, Statistics, Linear Algebra, Calculus, Database Management Systems, Algorithm Design and Analysis.
- **Self-study:** modern C++, CUDA, ML systems, causal inference.